

Module 3 review

Derivatives at a point: Rate of change of the function at that point/slope of the tangent to the function at that point.

Basic rules for finding the derivatives of a single variable function:

- **Trigonometry rules:**

1. When $f(x) = \sin(x)$, $f'(x) = \cos(x)$

2. When $f(x) = \cos(x)$, $f'(x) = -\sin(x)$

3. When $f(x) = \tan(x)$, $f'(x) = \sec^2(x)$

- **Power Rule:** When $f(x) = x^n$, $f'(x) = nx^{n-1}$

For example: $f(x) = x^4$, $f'(x) = 4x^3$

- **Multiplication by a constant:** When $f(x) = cg(x)$, $f'(x) = cg'(x)$

For example: $f(x) = 3x^4$, $f'(x) = 3 \cdot 4x^3 = 12x^3$

- **Sum/difference Rule:** When $f(x) = g(x) \pm h(x)$, $f'(x) = g'(x) \pm h'(x)$

For example: $f(x) = x^2 + x^3$, $f'(x) = 2x + 3x^2$

- **Product Rule:** When $f(x) = g(x)h(x)$, $f'(x) = g(x)h'(x) + h(x)g'(x)$

For example: $f(x) = \sin(x)\cos(x)$, $f'(x) = \sin(x)(-\sin(x)) + \cos(x)\cos(x) = \cos^2(x) - \sin^2(x)$

- **Quotient Rule:** When $f(x) = \frac{g(x)}{h(x)}$, $f'(x) = \frac{g'(x)h(x) - h'(x)g(x)}{h^2(x)}$

For example: $f(x) = \frac{\cos(x)}{x}$, $f'(x) = \frac{-\sin(x) \cdot x - 1 \cdot \cos(x)}{x^2} = -\frac{\cos(x) + x\sin(x)}{x^2}$

- **Reciprocal Rule:** When $f(x) = \frac{1}{g(x)}$, $f'(x) = \frac{-g'(x)}{g^2(x)}$

For example: $f(x) = \frac{1}{x}$, $f'(x) = \frac{-1}{x^2}$

- **Chain Rule:** When $f(x) = h(g(x))$, $f'(x) = h'(g(x))g'(x)$

For example: $f(x) = \sin(x^2)$, $f'(x) = \cos(x^2)2x = 2x \cos(x^2)$

Derivatives with respect to particular variables for multivariable functions

Since partial derivatives depend on the variable we are finding the derivative with respect to, the **notation** needs to be specific. For example, if we are finding the **partial derivative** of a function $f(x,y)$ with respect to x , then the notation is:

$$f_x(x, y) \text{ or } \frac{\partial f(x,y)}{\partial x}$$

Second Order Derivative

Second order derivatives are essentially **derivatives of derivatives**.

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} \text{ (2nd derivative in x direction)}$$

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} \text{ (derivative w.r.t x, then y)}$$

Critical points

Local maxima and minima of a function f are studied by looking for input values a where $f'(a) = 0$.

Such a point a has various names:

- **Stable point**
- **Critical point**
- **Stationary point**

How to find Local Maximum/Minimum points and Saddle points: Second order Derivative test

Suppose you have a function $f(x,y)$ and you found the critical point at (x_0, y_0) (using the first order derivative rule). We define D as:

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

- If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then (x_0, y_0) is a Local Maximum point and $f(x_0, y_0)$ is *Local Maximum*
- If $D > 0$ and $f_{xx}(x_0, y_0) > 0$ then (x_0, y_0) is a Local Minimum point and $f(x_0, y_0)$ is *Local Minimum*
- If $D < 0$ then (x_0, y_0) is a *Saddle Point*
- If $D = 0$, then the test is *inconclusive*

Gradient of a multivariable function

The gradient of a multivariable function $f(x,y,z,...)$ is a vector denoted by ∇f that packages all the partial derivatives of the function:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \\ \vdots \\ \vdots \end{bmatrix}$$

Directional Derivatives: Rate of change of a function in the direction of a given vector.

Notations:

- $\nabla_{\vec{v}} f$
- $\frac{\partial f}{\partial \vec{v}}$
- $f'_{\vec{v}}$

- $D_{\vec{v}}f$
- $\partial_{\vec{v}}f$

First step for finding the directional derivative is to find the *unit vector in the direction of $\vec{v}$* . We will call it $\vec{u}$.

$$\vec{u} = \frac{1}{\sqrt{v_1^2 + v_2^2}} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Second step is Finding the *dot product* of the gradient at point (x_0, y_0) and $\vec{u}$

$$\nabla_{\vec{u}} f(x_0, y_0) = \begin{bmatrix} \frac{\partial f(x_0, y_0)}{\partial x} \\ \frac{\partial f(x_0, y_0)}{\partial y} \end{bmatrix} \bullet \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \nabla f(x_0, y_0) \bullet \vec{u}$$

We get the *maximum rate of change (steepest ascent) when we move in the direction of the gradient of the function!*

Gradient descent

If we **start** at a point $\mathbf{x}_0$ and move a positive distance α in the direction of the negative gradient, then our new and improved $\mathbf{x}_1$ will look like this:

$$\mathbf{x}_1 = \mathbf{x}_0 - \alpha \nabla f(\mathbf{x}_0)$$

If we keep on taking our steps, then we can generalize:

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha \nabla f(\mathbf{x}_n)$$

Indefinite Integrals or Antiderivatives

Function	antiderivative
$f(x) = a$	$\int f(x)dx = ax + C$

$f(x) = ax^n$	$\int f(x)dx = \frac{ax^{n+1}}{n+1}$
$f(x) = ax^{-1}$	$\int f(x)dx = a \ln x + C$
$f(x) = ae^{kx}$	$\int f(x)dx = \frac{1}{k}ae^{kx} + C$
$f(x) = a \cos(kx)$	$\int f(x)dx = \frac{1}{k}a \sin(kx) + C$
$f(x) = a \sin(kx)$	$\int f(x)dx = -\frac{1}{k}a \cos(kx) + C$

Table 1: Some common antiderivatives (throughout a, k, C are constants)

Definite Integrals: Area under the curve between two points

Theorem: Let f be continuous on $[a, b]$, If F is an antiderivative of f then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Techniques for special cases

Preliminary Algebra

Method of Substitution

Works mostly when the integrand is a function of a function and a part of the integrand is a derivative of the inner function.

Integration by parts

$$\int u dv = uv - \int v du$$

Preferably choose that part of the integrand as u , which after differentiation, drops out x (the initial variable), reduces it maximum.

Let's Practice

1. A particle moves along a line with position $s(t) = t^3 - 6t^2 + 9t$ (units). For $t \geq 0$: find velocity & acceleration, times when particle is momentarily at rest.
2. The rate of water entering a tank is $R(t) = 5e^{-0.5t}$ L/min. Find the total water entering between $t = 0$ and $t = 5$.
3. Concentration of a pollutant is $C(x, y) = \frac{10}{1+x^2+y^2} \text{ mg/m}^3$. A sensor at $(1,0)$ wants to move to reduce concentration as fast as possible.
 - (a) Compute ∇C .
 - (b) At $(1,0)$, find the direction of fastest decrease as a unit vector, and the rate of decrease.
4. Let $f(x, y) = xe^y$. At point $(2,0)$:
 - (a) Find gradient and its magnitude.
 - (b) What is the maximum value of the directional derivative at that point, and in which direction does it occur?
5. Tree density along a side of a trail per square meter is

$$d(x) = 50 + 20e^{-0.2x}$$

where x is meters along a trail. If the trail is 10 meters long, How many trees are on both sides of the trail?

6. A farmer's land elevation is given by $h(x, y) = 10 + 3x - 2y + xy$ (meters). The farmer stands at $B = (2, 1)$. She wants to walk so that her elevation doesn't change to first order (i.e., stay on the same level set instantaneously). Find a unit direction vector she can walk in to keep elevation unchanged initially.
7. A bacterial colony growth rate (cells per hour) is modeled by:

$$r(t) = 200e^{-0.1t}, 0 \leq t \leq 24.$$

- How many new bacteria are produced in the first 24 hours?
- What fraction of total growth occurs in the first 6 hours?

Solutions:

1. velocity = $s'(t) = \frac{d(t^3 - 6t^2 + 9t)}{dt} = 3t^2 - 12t + 9$

acceleration = $s''(t) = \frac{d(3t^2 - 12t + 9)}{dt} = 6t - 12$

2. given,
 $R(t) = 5e^{-0.5t} \text{ L/min}$

∴ Total water entering between $t=0$ to $t=5$

$$\begin{aligned} &= \int_0^5 5e^{-0.5t} dt = \left. \frac{-5}{0.5} e^{-0.5t} \right|_0^5 = \left. -10e^{-0.5t} \right|_0^5 \\ &= \left(-10e^{(-0.5 \times 5)} \right) - \left(-10e^{(-0.5 \times 0)} \right) \\ &= -10e^{-2.5} + 10e^0 = -10e^{-2.5} + 10 \\ &= 10(1 - e^{-2.5}) \text{ L} \\ &\approx 9.18 \text{ liters} \end{aligned}$$

3. Given,

$$C(x, y) = \frac{10}{1+x^2+y^2}$$

$$a) \nabla C(x, y) = \begin{bmatrix} \frac{-20x}{(1+x^2+y^2)^2} \\ \frac{-20y}{(1+x^2+y^2)^2} \end{bmatrix}$$

$$b) \nabla C(1, 0) = \begin{bmatrix} \frac{-20}{4} \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \end{bmatrix}$$

Direction of fastest decrease is -ve of the gradient: $-\nabla C(1, 0) = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$

Unit vector in the direction, $\vec{u} = \begin{bmatrix} \frac{5}{\sqrt{5^2+0^2}} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Rate of fastest decrease: Directional derivative in the direction of $\vec{u} = \begin{bmatrix} -5 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -5$

$\therefore$ Rate: -5 mg/m^3

$$4. \quad \nabla f(x, y) = \begin{bmatrix} e^y \\ x e^y \end{bmatrix}$$

$$\nabla f(2, 0) = \begin{bmatrix} e^0 \\ 2 \cdot e^0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{magnitude of gradient} = \sqrt{1^2 + 2^2} = \sqrt{5}$$

Max. directional derivative : In the direction of the gradient

→ unit vector in the direction of $\nabla f(2, 0)$,

$$\vec{u} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\therefore \nabla f(2, 0) \cdot \vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} + \frac{4}{\sqrt{5}} = \frac{5}{\sqrt{5}} = \sqrt{5}$$

(you can directly find $|\nabla f(2, 0)|$ also)

5. True on 1 side:

$$\begin{aligned} & \int_0^{10} (50 + 20e^{-0.2x}) dx \\ &= \left[50x + \frac{20}{-0.2} e^{-0.2x} \right]_0^{10} \\ &= 500 - \frac{20}{0.2} e^{(-0.2 \times 10)} - 0 + \frac{20}{0.2} e^0 \\ &= 600 - 100e^{-2} \end{aligned}$$

True on BOTH sides:

$$\begin{aligned} & 2 \times (600 - 100e^{-2}) \\ &= 1200 - 200e^{-2} \approx 1172.94 \approx 1173 \text{ True} \end{aligned}$$

$$6. \nabla h = \begin{bmatrix} 3+y \\ -2+x \end{bmatrix}$$

$$\text{at } B = (2,1), \nabla h(2,1) = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

To stay on the same level, directional derivative needs to be 0. Let $\vec{v}$ be the direction in which she walks to keep her level the same

$$\therefore \begin{bmatrix} 4 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \quad \Rightarrow \quad 4v_1 = 0$$

$$\Rightarrow v_1 = 0$$

v_2 can be anything. $\therefore$ To generalize, we can say any vector in the direction of the unit vector $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

7. Given,
growth rate $\Rightarrow r(t) = 200e^{-0.1t}$ cells/hr where
 $0 \leq t \leq 24$

a) $\therefore$ Total growth in 24 hrs:

$$\int_0^{24} 200e^{-0.1t} dt = \frac{200}{-0.1} e^{-0.1t} \Big|_0^{24}$$

$$= 2000(1 - e^{-2.4}) \approx 1818.6 \approx 1819$$

b) Fraction of growth in 1st six hours:

= growth in 1st 6 hrs

$$= \frac{\int_0^6 200e^{-0.1t} dt}{2000(1 - e^{-2.4})} = \frac{2000(1 - e^{-0.6})}{2000(1 - e^{-2.4})}$$
$$\approx 49.6\%$$